MATHEMATICS PART 1 (ALGEBRA) | CLASS 10 | MAHARASHTRA SSC BOARD

Chapter 5: Probability

MODEL ANSWER SHEET / ANSWER KEY

40	12	1
QUESTIONS	IN TEXTBOOK	FIGURES

HOW TO USE
Cover the answer box ✓ and answer first. A **TEXTBOOK** *strip means the same question is in your Balbharati Math-1 textbook, at the set and page shown. Tick ☐ each section once revised.*

Q.1 (A)	Q.1 (B)	Q.2 (A)	Q.2 (B)	Q.3 (A)	Q.3 (B)	Q.4	Q.5
Multiple Choice	One-step Solve	Activities	Short Solve	Activities	Solve / Prove	Long Solve / Prove	Creative / HOTS
7Q	5Q	4Q	5Q	1Q	8Q	5Q	5Q

Q.1 (A) Multiple Choice — 1 Mark each

7 QUESTIONS · 2 ☐ Revised

Q1 Which of the following number cannot represent a probability?

(a) 0.66 (b) 1.5 (c) 0.15 (d) 0.7

ANSWER / SOLUTION

1.5

Explanation:

The probability of any event is from 0 to 1.

Q2 If $n(A) = 5$, $P(A) = \frac{1}{2}$, then $n(S) = ?$

(a) 10 (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) $\frac{1}{3}$

ANSWER / SOLUTION

10

Explanation:

$P(A) = \frac{n(A)}{n(S)}$

$\therefore \frac{1}{2} = \frac{5}{n(S)}$

$\therefore n(S) = 10$

Q3 When a dice is thrown, the number of sample points in the sample space are _____.

(a) 4 (b) 6 (c) 2 (d) 52

ANSWER / SOLUTION

When a dice is thrown, the number of sample points in the sample space are **6**.

Explanation:

$S = \{1, 2, 3, 4, 5, 6\}$

$\therefore n(S) = 6$

Q4 In how many ways a card can be drawn from a well shuffled pack of playing cards?

TEXTBOOK **Practice Set 5.1 · Q.1 (3) · Page 116**

(a) 4 (b) 1 (c) 26 (d) 52

ANSWER / SOLUTION

52

Explanation:

A card can be drawn from 52 playing cards in 52 ways.

Q5 What is the probability of the event that a number chosen from 1 to 50 is a prime number?

(a) $\frac{3}{10}$ (b) $\frac{1}{2}$
(c) $\frac{1}{4}$ (d) $\frac{3}{25}$

ANSWER / SOLUTION

$\frac{3}{10}$

Explanation:

$S = \{1, 2, \ldots, 50\}$

$\therefore n(S) = 50$

Let A be the event of getting a prime number.

$\therefore A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$

$\therefore n(A) = 15$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{15}{50} = \frac{3}{10}$

Q6 Which of the following options shows the highest probability?

TEXTBOOK **Problem Set 5 · Q.2 · Page 126**

(a) $\frac{4}{5}$ (b) 0.83 (c) 58% (d) $\frac{1}{2}$

ANSWER / SOLUTION

0.83

Explanation:

Convert each option to the same form (decimals or percentages):

A) $\frac{4}{5} = 0.80 \quad (80\%)$

B) $0.83 = 0.83 \quad (83\%)$

C) $58\% = \frac{58}{100} = 0.58 \quad (58\%)$

D) $\frac{1}{2} = 0.50 \quad (50\%)$

0.83 is the largest.

Q7 When two dice are thrown, the number of sample points in the sample space are _____.

(a) 6 (b) 12 (c) 36 (d) 52

ANSWER / SOLUTION

When two dice are thrown, the number of sample points in the sample space are **36**.

Explanation:

When two dice are thrown,

$n(S) = 6 \times 6 = 36$

Q.1 (B) One-step Solve 1 Mark each

5 QUESTIONS · 1 ☐ Revised

Q1 Write a sample space if two coins are tossed simultaneously.

ANSWER / SOLUTION

When two coins are tossed,

$S = \{\mathrm{HH}, \mathrm{HT}, \mathrm{TH}, \mathrm{TT}\}$

Q2 Write a sample space when a die is thrown.

ANSWER / SOLUTION

Sample space for one die thrown,

$S = \{1, 2, 3, 4, 5, 6\}.$

Q3 In a set of 25 cards, each card bears only one number from 1 to 25. One card is drawn randomly. Write the sample space for this random experiment.

ANSWER / SOLUTION

$S = \{1, 2, 3, \ldots, 25\}$

Q4 A two-digit number is formed with digits $2, 3, 5$ without repetition. Write the sample space.

TEXTBOOK **Practice Set 5.2 · Q.1 (2) · Page 117**

ANSWER / SOLUTION

$S = \{23, 25, 32, 35, 52, 53\}$

$n(S) = 6$

Q5 Write the event in the set form for the following random experiment. 'If one die is thrown, the number obtained on the upper face is even.'

ANSWER / SOLUTION

$S = \{1, 2, 3, 4, 5, 6\}$

Let A be the event that the number obtained on the upper face is even.

$\therefore A = \{2, 4, 6\}$

Q.2 (A) Activities 2 Marks each

4 QUESTIONS · 2 ☐ Revised

Q1 If one die is rolled, then find the probability of the following event by completing the activity.

Event A: Number on the upper face is prime.

Activity: Let 'S' be the sample space.

$S = \{1, 2, 3, 4, 5, 6\}$

$\therefore n(S) = 6$

Event A: Prime number on the upper face.

$A = \{\square\}$

$\therefore n(A) = 3$

$P(A) = \frac{\square}{n(S)}$ ---[Formula]

$= \frac{\square}{6}$

$\therefore P(A) = \frac{1}{\square}$

TEXTBOOK **Solved Example** · Ex. (2) (i) · **Page 123**

ANSWER / SOLUTION

Let 'S' be the sample space.

$S = \{1, 2, 3, 4, 5, 6\}$

$\therefore n(S) = 6$

Event A: Prime number on the upper face.

$A = \{\boxed{2, 3, 5}\}$

$\therefore n(A) = 3$

$P(A) = \frac{\boxed{n(A)}}{n(S)}$ ---[Formula]

$= \frac{\boxed{3}}{6}$

$\therefore P(A) = \frac{1}{\boxed{2}}$

Q2 Two coins are tossed simultaneously. Write the sample space (S) and expected sample points in the given events by completing the activity.

i. Event A: to get at least one head.

ii. Event B: to get no head.

Activity: Let 'S' be the sample space, when two coins are tossed simultaneously.

$\therefore S = \{\square, \text{HT}, \text{TH}, \square\}$

Event A: to get at least one head.

$\therefore A = \{\text{HH}, \square, \text{TH}\}$

Event B: to get no head.

$\therefore B = \{\square\}$

ANSWER / SOLUTION

Activity: Let 'S' be the sample space, when two coins are tossed simultaneously.

$\therefore S = \{\boxed{\text{HH}}, \text{HT}, \text{TH}, \boxed{\text{TT}}\}$

Event A: to get at least one head.

$\therefore A = \{\text{HH}, \boxed{\text{HT}}, \text{TH}\}$

Event B: to get no head.

$\therefore B = \{\boxed{\text{TT}}\}$

Q3 **A card is drawn from a well shuffled pack of 52 playing cards. Find the probability of the event, the card drawn is a red card.**

Activity: Let 'S' be the sample space.

$\therefore n(S) = 52$

Event A: Card drawn is a red card.

$\therefore$ Total red cards $= \square$ hearts $+ 13$ diamonds

$\therefore n(A) = \square$

$\therefore P(A) = \frac{\square}{n(S)}$ ---[Formula]

$P(A) = \frac{26}{52}$

$P(A) = \square$

TEXTBOOK **Solved Example · Ex. (3) (i) · Page 124**

ANSWER / SOLUTION

Let 'S' be the sample space.

$\therefore n(S) = 52$

Event A: Card drawn is a red card.

$\therefore$ Total red cards $= \boxed{13}$ hearts $+ 13$ diamonds

$\therefore n(A) = \boxed{26}$

$\therefore P(A) = \frac{\boxed{n(A)}}{n(S)}$ ---[Formula]

$P(A) = \frac{26}{52}$

$P(A) = \boxed{\frac{1}{2}}$

Q4 **In Adarsh High School, out of 30 students in a class 3 students wear glasses (spectacles). If a student in the class is randomly selected, find the probability that he or she wears glasses (spectacles) by completing the following activity.**

Activity: There are a total of 30 students in the class.

$\therefore n(S) = \square$

Event A: selected student wears glasses (spectacles).

$\therefore n(A) = \square$

$P(A) = \frac{\square}{n(S)}$ ---[Formula]

$P(A) = \square$

ANSWER / SOLUTION

There are a total of 30 students in the class.

$\therefore n(S) = \boxed{30}$

Event A: selected student wears glasses (spectacles).

$\therefore n(A) = \boxed{3}$

$P(A) = \frac{\boxed{n(A)}}{n(S)}$ ---[Formula]

$P(A) = \frac{3}{30} = \boxed{\frac{1}{10}}$

Q.2 (B) Short Solve 2 Marks each

5 QUESTIONS · 1 ☐ Revised

Q1 A card is drawn at random from a pack of well shuffled 52 playing cards. Find the probability that the card drawn is a spade.

TEXTBOOK Practice Set 5.4 · Q.5 (2) · Page 125

ANSWER / SOLUTION

There are 52 playing cards.

$\therefore n(S) = 52$

Let B be the event that the card drawn is a spade.

There are 13 spade cards.

$\therefore n(B) = 13$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{13}{52} = \frac{1}{4}$

Hence, the probability that the card drawn is a spade is $\frac{1}{4}$.

Q2 If two coins are tossed, find the probability of event getting head on both the coins.

ANSWER / SOLUTION

Sample space,

$S = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}$

$\therefore n(S) = 4$

Let A be the event of getting head on both the coins.

$\therefore A = \{\text{HH}\}$

$\therefore n(A) = 1$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{1}{4}$

Q3 If one die is rolled, then find the probability of event that the number on the upper face is greater than 6?

ANSWER / SOLUTION

Sample space,

$S = \{1, 2, 3, 4, 5, 6\}$

$\therefore n(S) = 6$

Let A be the event that the number on the upper face is greater than 6.

Here, the greatest number is 6.

$\therefore$ Event A is an impossible event.

$\therefore A = \{\}$

$\therefore n(A) = 0$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{0}{6} = 0$

Q4 If three coins are tossed simultaneously, find the probability of the event to get no head.

ANSWER / SOLUTION

Sample space,

$S = \{\text{HHH}, \text{HHT}, \text{HTH}, \text{THH}, \text{HTT}, \text{THT}, \text{TTH}, \text{TTT}\}$

$\therefore n(S) = 8$

Let A be the event of getting no head.

$\therefore A = \{\text{TTT}\}$

$\therefore n(A) = 1$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{1}{8}$

Q5 There are 30 cards in a box, each bearing one of the numbers from 1 to 30. One card is drawn at random from the box. Find the probability of event that the card drawn shows a number which is a multiple of 5.

ANSWER / SOLUTION

Sample space,

$S = \{1, 2, 3, \ldots, 30\}$

$\therefore n(S) = 30$

Let A be the event that the card drawn shows a number which is a multiple of 5.

$\therefore A = \{5, 10, 15, 20, 25, 30\}$

$\therefore n(A) = 6$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{6}{30} = \frac{1}{5}$

Q.3 (A) Activities

3 Marks each

1 QUESTIONS · 1 · Revised

Q1 A box contains 5 strawberry chocolates, 6 coffee chocolates and 2 peppermint chocolates. If one of the chocolates is picked from the box at random, find the probability of the following events by completing the activity.

i. Event A: it is a coffee chocolate.

ii. Event B: it is a peppermint chocolate.

Activity: Let 'S' be the sample space.

$\therefore n(S) = 5 + 6 + 2 = 13$

i. Event A: it is a coffee chocolate.

$\therefore n(A) = \square$

$\therefore P(A) = \frac{\square}{n(S)}$ ---[Formula]

$P(A) = \frac{\square}{13}$

ii. Event B: it is a peppermint chocolate.

$\therefore n(B) = \square$

$\therefore P(B) = \frac{\square}{n(S)}$ ---[Formula]

$P(B) = \frac{\square}{13}$

TEXTBOOK **Solved Example** · Ex. (4) · **Page 124**

ANSWER / SOLUTION

Let 'S' be the sample space.

$\therefore n(S) = 5 + 6 + 2 = 13$

i. Event A: it is a coffee chocolate.

$\therefore n(A) = \boxed{6}$

$\therefore P(A) = \frac{\boxed{n(A)}}{n(S)}$ ---[Formula]

$P(A) = \frac{\boxed{6}}{13}$

ii. Event B: it is a peppermint chocolate.

$\therefore n(B) = \boxed{2}$

$\therefore P(B) = \frac{\boxed{n(B)}}{n(S)}$ ---[Formula]

$P(B) = \frac{\boxed{2}}{13}$

Q.3 (B) Solve / Prove 3 Marks each

8 QUESTIONS · 4 ☐ Revised

Q1 If two dice are rolled simultaneously, find the probability of the following events.

Event A: The sum of the digits on the upper faces is at least 10.

TEXTBOOK Practice Set 5.4 · Q.2 (1) · Page 125

ANSWER / SOLUTION

Sample space,

$S = \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),$
$(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),$
$(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),$
$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),$
$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),$
$(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$

$\therefore n(S) = 36$

Let A be the event that the sum of the digits on the upper faces is at least 10.

$\therefore A = \{(4,6),(5,5),(5,6),(6,4),(6,5),(6,6)\}$

$\therefore n(A) = 6$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

Q2 If two dice are rolled simultaneously, find the probability of the following events.

Event B: The sum of the digits on the upper faces is 33.

TEXTBOOK Practice Set 5.4 · Q.2 (2) · Page 125

ANSWER / SOLUTION

Sample space,

$S = \{(1,1),\ldots,(6,6)\}$

$\therefore n(S) = 36$

Let B be the event that the sum of the digits on the upper faces is 33.

The sum of the digits on the upper faces can be maximum 12.

$\therefore$ Event B is an impossible event.

$\therefore B = \{\}$

$\therefore n(B) = 0$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{0}{36} = 0$

Q3 If three coins are tossed simultaneously, find the probability of the following events.

Event A: To get no head.

ANSWER / SOLUTION

Sample space,

$S = \{\text{HHH},\text{HHT},\text{HTH},\text{THH},\text{HTT},\text{THT},\text{TTH},\text{TTT}\}$

$\therefore n(S) = 8$

Let A be the event of getting no head.

$\therefore A = \{\text{TTT}\}$

$\therefore n(A) = 1$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{1}{8}$

Q4 If three coins are tossed simultaneously, find the probability of the following events.

Event B: To get at least two heads.

✓ ANSWER / SOLUTION

Sample space,

$S = \{\text{HHH}, \text{HHT}, \text{HTH}, \text{THH}, \text{HTT}, \text{THT}, \text{TTH}, \text{TTT}\}$

$\therefore n(S) = 8$

Let B be the event of getting at least two heads.

$\therefore B = \{\text{HHH}, \text{HHT}, \text{HTH}, \text{THH}\}$

$\therefore n(B) = 4$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{4}{8} = \frac{1}{2}$

Q5 If one coin and one die are thrown simultaneously, find the probability of the following events.

Event A: To get a tail and an even number.

✓ ANSWER / SOLUTION

Sample space,

$S = \{(\text{H}, 1), (\text{H}, 2), (\text{H}, 3), (\text{H}, 4), (\text{H}, 5), (\text{H}, 6), (\text{T}, 1), (\text{T}, 2), (\text{T}, 3), (\text{T}, 4), (\text{T}, 5), (\text{T}, 6)\}$

$\therefore n(S) = 12$

Event A: To get a tail and an even number.

$\therefore A = \{(\text{T}, 2), (\text{T}, 4), (\text{T}, 6)\}$

$\therefore n(A) = 3$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{3}{12} = \frac{1}{4}$

Q6 If one coin and one die are thrown simultaneously, find the probability of the following events.

Event B: To get head and an odd number.

✓ ANSWER / SOLUTION

Sample space,

$S = \{(\text{H}, 1), (\text{H}, 2), (\text{H}, 3), (\text{H}, 4), (\text{H}, 5), (\text{H}, 6), (\text{T}, 1), (\text{T}, 2), (\text{T}, 3), (\text{T}, 4), (\text{T}, 5), (\text{T}, 6)\}$

$\therefore n(S) = 12$

Event B: To get head and an odd number.

$\therefore B = \{(\text{H}, 1), (\text{H}, 3), (\text{H}, 5)\}$

$\therefore n(B) = 3$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{3}{12} = \frac{1}{4}$

Q7 A balloon vendor has 2 red, 3 blue and 4 green balloons. He wants to choose one of them at random to give it to Pranali. What is the probability of the event that Pranali gets, a red balloon.

📘 **TEXTBOOK** **Problem Set 5** · Q.5 (1) · **Page 126**

✓ ANSWER / SOLUTION

Given: Number of red balloon $= 2$

Number of blue balloon $= 3$

Number of green balloon $= 4$

Let S be the sample space.

$\therefore n(S) = \text{Red} + \text{Blue} + \text{Green} = 2 + 3 + 4 = 9$

Let A be the event of getting a red balloon.

Given: $n(A) = 2$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{2}{9}$

Q7 *continued...*

ANSWER / SOLUTION (CONTD.)

Hence, the probability of the event that Pranali gets a red balloon is $\frac{2}{9}$.

SOLUTION 2

Let the 2 red balloons be R_1, R_2, 3 blue balloons be B_1, B_2, B_3 and 4 green balloons be G_1, G_2, G_3, G_4.

$\therefore$ Sample space, $S = \{R_1, R_2, B_1, B_2, B_3, G_1, G_2, G_3, G_4\}$

$\therefore n(S) = 9$

Let A be the event that Pranali gets a red balloon.

$\therefore A = \{R_1, R_2\}$

$\therefore n(A) = 2$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{2}{9}$

$\therefore$ The probability that Pranali gets a red balloon is $\frac{2}{9}$.

Q8 **A balloon vendor has 2 red, 3 blue and 4 green balloons. He wants to choose one of them at random to give it to Pranali. What is the probability of the event that Pranali gets, a blue balloon.**

TEXTBOOK **Problem Set 5 · Q.5 (2) · Page 126**

ANSWER / SOLUTION

Given: Number of red balloon $= 2$

Number of blue balloon $= 3$

Number of green balloon $= 4$

Let S be the sample space.

$\therefore n(S) = \text{Red} + \text{Blue} + \text{Green} = 2 + 3 + 4 = 9$

Let B be the event of getting a blue balloon.

Given: $n(B) = 3$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{3}{9} = \frac{1}{3}$

Hence, the probability of the event that Pranali gets a blue balloon is $\frac{1}{3}$.

SOLUTION 2

Let the 2 red balloons be R_1, R_2, 3 blue balloons be B_1, B_2, B_3 and 4 green balloons be G_1, G_2, G_3, G_4.

$\therefore$ Sample space, $S = \{R_1, R_2, B_1, B_2, B_3, G_1, G_2, G_3, G_4\}$

$\therefore n(S) = 9$

Let B be the event that Pranali gets a blue balloon.

$\therefore B = \{B_1, B_2, B_3\}$

$\therefore n(B) = 3$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{3}{9} = \frac{1}{3}$

$\therefore$ The probability that Pranali gets a blue balloon is $\frac{1}{3}$.

Q.4 Long Solve / Prove 4 Marks each

5 QUESTIONS ☐ Revised

Q1 What is the probability that an ordinary year has 53 Sundays?

ANSWER / SOLUTION

Ordinary year has 365 days.

$365 \text{ days} = 52 \text{ weeks} + 1 \text{ day}$

That 1 day may be Sun, Mon, Tue, Wed, Thu, Fri, Sat.

Total no. of possible outcomes $= 7$

Let $E \longrightarrow$ event of getting 53 Sundays.

No. of favourable outcomes $= 1 \quad \{\text{Sun}\}$

$P(E) = \frac{\text{No. of favorable outcomes}}{\text{Total no. of possible outcomes}} = \frac{1}{7}$

SOLUTION 2

There are 365 days i.e., 52 weeks and 1 day in an ordinary year.

For the remaining day, the sample space is

$S = \{\text{Mon.}, \text{Tue.}, \text{Wed.}, \text{Thur.}, \text{Fri.}, \text{Sat.}, \text{Sun.}\}$

$\therefore n(S) = 7$

Let A be the event that the remaining day is a Sunday.

$\therefore A = \{\text{Sunday}\}$

$\therefore n(A) = 1$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{1}{7}$

$\therefore$ The probability that an ordinary year has 53 Sundays is $\frac{1}{7}$.

Q2 What is the probability that a leap year has 53 Sundays?

ANSWER / SOLUTION

A leap year has 366 days i.e., 52 weeks and 2 days.

There are 52 Sundays in 52 weeks.

For the remaining 2 days, sample space is:

$S = \{(\text{Mon.}, \text{Tue.}), (\text{Tue.}, \text{Wed.}), (\text{Wed.}, \text{Thur.}), (\text{Thur.}, \text{Fri.}), (\text{Fri.}, \text{Sat.}), (\text{Sat.}, \text{Sun.}), (\text{Sun.}, \text{Mon.})\}$

$\therefore n(S) = 7$

Let A be the event that the remaining two days are Sundays.

$\therefore A = \{(\text{Sat.}, \text{Sun.}), (\text{Sun.}, \text{Mon.})\}$

$\therefore n(A) = 2$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{2}{7}$

$\therefore$ The probability that a leap year has 53 Sundays is $\frac{2}{7}$.

Q3 A box contains 36 cards, bearing only one number from 1 to 36 on each. If one card is drawn at random, find the probability of an event that the card drawn bears, a complete square number.

ANSWER / SOLUTION

Sample space,

$S = \{1, 2, 3, \ldots, 36\}$

$\therefore n(S) = 36$

Let A be the event that the card drawn bears a complete square number.

$\therefore A = \{1, 4, 9, 16, 25, 36\}$

$\therefore n(A) = 6$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

Q4 **A box contains 36 cards, bearing only one number from 1 to 36 on each. If one card is drawn at random, find the probability of an event that the card drawn bears, a prime number.**

✅ **ANSWER / SOLUTION**

Sample space,

$S = \{1, 2, 3, \ldots, 36\}$

$\therefore n(S) = 36$

Let B be the event that the card drawn bears a prime number.

$\therefore B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31\}$

$\therefore n(B) = 11$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{11}{36}$

Q5 **A box contains 36 cards, bearing only one number from 1 to 36 on each. If one card is drawn at random, find the probability of an event that the card drawn bears, a number divisible by 3.**

✅ **ANSWER / SOLUTION**

Sample space,

$S = \{1, 2, 3, \ldots, 36\}$

$\therefore n(S) = 36$

Let C be the event that the card drawn bears a number divisible by 3.

$\therefore C = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36\}$

$\therefore n(C) = 12$

$\therefore P(C) = \frac{n(C)}{n(S)} = \frac{12}{36} = \frac{1}{3}$

Q.5 🚀 Creative / HOTS 3 Marks each

5 QUESTIONS · 📘 1 ☐ Revised

Q1 **A bag contains 5 white balls and some blue balls. If the probability of drawing a blue ball is double that of a white ball, determine the number of blue balls in the bag.**

✅ **ANSWER / SOLUTION**

Let the number of blue balls be x.

Number of white balls $= 5$

$\therefore$ Total number of balls $= (x + 5)$

$P(\text{blue ball is drawn}) = \frac{x}{x+5}$

$P(\text{white ball is drawn}) = \frac{5}{x+5}$

According to the given condition, the probability of drawing a blue ball is double that of a white ball.

$\therefore P(\text{blue ball is drawn}) = 2 \times P(\text{white ball is drawn})$

$\therefore \frac{x}{x+5} = 2 \times \frac{5}{x+5}$

$\therefore x(x + 5) = 10(x + 5)$

$\therefore x^2 + 5x = 10x + 50$

$\therefore x^2 - 5x - 50 = 0$

$\therefore x^2 - 10x + 5x - 50 = 0$

$\therefore x(x - 10) + 5(x - 10) = 0$

$\therefore (x - 10)(x + 5) = 0$

$\therefore x - 10 = 0$ or $x + 5 = 0$

$\therefore x = 10$ or $x = -5$

But, number of balls cannot be negative.

$\therefore x = 10$

$\therefore$ The number of blue balls in the bag is 10.

Q2 **The faces of a die bear numbers $0, 1, 2, 3, 4, 5$. If the die is rolled twice, then find the probability that the product of digits on the upper face is zero.**

TEXTBOOK **Problem Set 5** · Q.16 · **Page 128**

ANSWER / SOLUTION

Sample space,

$S = \{(0,0), (0,1), (0,2), (0,3), (0,4), (0,5),$
$(1,0), (1,1), (1,2), (1,3), (1,4), (1,5),$
$(2,0), (2,1), (2,2), (2,3), (2,4), (2,5),$
$(3,0), (3,1), (3,2), (3,3), (3,4), (3,5),$
$(4,0), (4,1), (4,2), (4,3), (4,4), (4,5),$
$(5,0), (5,1), (5,2), (5,3), (5,4), (5,5)\}$

$\therefore n(S) = 36$

Let A be the event that the product of digits on the upper face is zero.

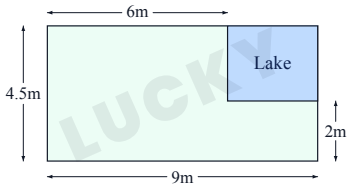
$\therefore A = \{(0,0), (0,1), (0,2), (0,3), (0,4), (0,5), (1,0), (2,0), (3,0), (4,0), (5,0)\}$

$\therefore n(A) = 11$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{11}{36}$

$\therefore$ The probability that the product of the digits on the upper face is zero is $\frac{11}{36}$.

Q3 **A missing helicopter is reported to have crashed somewhere in the rectangular region shown in the figure. What is the probability that it crashed inside the lake shown in the figure?**



ANSWER / SOLUTION

The helicopter is equally likely to crash anywhere in the region.

In the figure, the length and the breadth of the rectangle are $9\ \text{m}$ and $4.5\ \text{m}$ respectively.

Area of the entire region where the helicopter can crash $= (9 \times 4.5)\ \text{m}^2 = 40.5\ \text{m}^2$

Let A be the event that helicopter crashed inside the lake.

The lake is rectangular shaped.

Length of lake $= 9 - 6 = 3\ \text{m}$

Breadth of lake $= 4.5 - 2 = 2.5\ \text{m}$

Area of lake $= \text{length} \times \text{breadth} = 3 \times 2.5 = 7.5\ \text{m}^2$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{7.5}{40.5} = \frac{75}{405} = \frac{5}{27}$

The probability that the helicopter crashed inside the lake is $\frac{5}{27}$.

Q4 **Three horses A, B and C are in a race, A is twice as likely to win as B and B is twice as likely to win as C. What are their probabilities of winning?**

✅ **ANSWER / SOLUTION**

Let $P(C)$ be x.

Probability of horse B winning the race is twice as likely as C.

$\therefore P(B) = 2 \times P(C) = 2x$

Probability of horse A winning the race is twice as likely as B.

$P(A) = 2 \times P(B) = 2 \times 2x = 4x$

The total probability is 1.

$\therefore P(A) + P(B) + P(C) = 1$

$\therefore 4x + 2x + x = 1 \implies 7x = 1 \implies x = \frac{1}{7}$

$\therefore P(C) = x = \frac{1}{7}$

$P(B) = 2x = 2 \times \frac{1}{7} = \frac{2}{7}$

$P(A) = 4x = 4 \times \frac{1}{7} = \frac{4}{7}$

$\therefore P(A) = \frac{4}{7}, P(B) = \frac{2}{7}, P(C) = \frac{1}{7}$

Q5 **A bag contains 8 red balls and some blue balls. If one ball is drawn randomly the probability of drawing a red ball to a blue ball are in the ratio $2:5$, determine the probability of drawing a blue ball from the bag.**

✎ **CORRECTED** ~~red ball to a blue ball are in the ratio~~ $5:2$ → **red ball to a blue ball are in the ratio** $2:5$ — ratio reversed: with 5 : 2 blue balls = 3.2 (impossible); the solution uses red : blue = 2 : 5

✅ **ANSWER / SOLUTION**

Let the number of blue balls be x.

Number of red balls $= 8$

$\therefore$ Total number of balls $= (x + 8)$

$P(\text{blue ball is drawn}) = \frac{x}{x+8}$

$P(\text{red ball is drawn}) = \frac{8}{x+8}$

According to the given condition, the probability of drawing a blue ball to a red ball are in the ratio $5:2$.

$\therefore \frac{P(\text{blue ball is drawn})}{P(\text{red ball is drawn})} = \frac{5}{2}$

$\therefore \frac{\frac{x}{x+8}}{\frac{8}{x+8}} = \frac{5}{2}$

$\therefore \frac{x}{8} = \frac{5}{2} \implies 2x = 40 \implies x = 20$

But, number of balls cannot be negative.

$\therefore x = 20$

$\therefore P(\text{blue ball is drawn}) = \frac{x}{x+8} = \frac{20}{20+8} = \frac{20}{28} = \frac{5}{7}$

$\therefore$ The probability that a blue ball is drawn from the bag is $\frac{5}{7}$.

Practice daily. Solve every step. Score full marks. ⚡
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